

DAY-5

NETWORK TESTING

1. Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other. On Windows, use 'ping [IP address]' in Command Prompt; on Android, try a ping app or use Termux .

Setup: A windows laptop and a friends laptop were connect to the same home wi-fi network.

Command executed on laptop : ping 192.168.1.62

```
C:\Users\Admin>ping 192.168.1.62

Pinging 192.168.1.62 with 32 bytes of data:
Reply from 192.168.1.62: bytes=32 time=314ms TTL=128
Reply from 192.168.1.62: bytes=32 time=168ms TTL=128
Reply from 192.168.1.62: bytes=32 time=42ms TTL=128
Reply from 192.168.1.62: bytes=32 time=5ms TTL=128

Ping statistics for 192.168.1.62:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 5ms, Maximum = 314ms, Average = 132ms

C:\Users\Admin>|
```

2. Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.

Command : tracert www.youtube.com

```
C:\Users\Admin>tracert www.youtube.com

Tracing route to www.youtube.com [2001:4860:482d:400::]
over a maximum of 30 hops:

  1     3 ms     2 ms     2 ms    2401:4900:8fed:2bf7:1633:75ff:feed:45b3
  2     *        *        *      Request timed out.
  3    25 ms     5 ms     5 ms    fc00::83
  4    11 ms     7 ms     5 ms    2404:a800:2a00:103::2d
  5    53 ms    101 ms    102 ms   2404:a800::167
  6    99 ms    102 ms    98 ms    2001:4860:482d:400::

Trace complete.
```

3. Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.

Action: the friends laptop was disconnected from the wi-fi

Command executed : ping 192.168.1.46

```

Microsoft Windows [Version 10.0.26200.8875]
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C:\Users\Admin>ping 192.168.1.62

Pinging 192.168.1.62 with 32 bytes of data:
Request timed out.
Request timed out.
Reply from 192.168.1.123: Destination host unreachable.
Reply from 192.168.1.123: Destination host unreachable.

Ping statistics for 192.168.1.62:
    Packets: Sent = 4, Received = 2, Lost = 2 (50% loss),

```

4. Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.

```

C:\Users\Admin>arp -a

Interface: 192.168.37.1 --- 0x6
    Internet Address      Physical Address      Type
    192.168.37.254        00-50-56-f8-57-f1    dynamic
    192.168.37.255        ff-ff-ff-ff-ff-ff    static
    224.0.0.2             01-00-5e-00-00-02    static
    224.0.0.22            01-00-5e-00-00-16    static
    224.0.0.251           01-00-5e-00-00-fb    static
    224.0.0.252           01-00-5e-00-00-fc    static
    224.0.0.253           01-00-5e-00-00-fd    static
    224.77.77.77          01-00-5e-4d-4d-4d    static
    239.254.127.63        01-00-5e-7e-7f-3f    static
    239.255.102.18        01-00-5e-7f-66-12    static
    239.255.255.250       01-00-5e-7f-ff-fa    static
    239.255.255.254       01-00-5e-7f-ff-fe    static
    255.255.255.255       ff-ff-ff-ff-ff-ff    static

Interface: 192.168.1.123 --- 0x8
    Internet Address      Physical Address      Type
    192.168.1.1           14-33-75-ed-45-b3    dynamic
    224.0.0.2             01-00-5e-00-00-02    static
    224.0.0.22            01-00-5e-00-00-16    static
    224.0.0.251           01-00-5e-00-00-fb    static
    224.0.0.252           01-00-5e-00-00-fc    static
    224.0.0.253           01-00-5e-00-00-fd    static
    224.77.77.77          01-00-5e-4d-4d-4d    static
    239.254.127.63        01-00-5e-7e-7f-3f    static
    239.255.102.18        01-00-5e-7f-66-12    static
    239.255.255.250       01-00-5e-7f-ff-fa    static
    239.255.255.254       01-00-5e-7f-ff-fe    static
    255.255.255.255       ff-ff-ff-ff-ff-ff    static

```

5. Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed. Include screenshots or command outputs if possible.

I connected the laptop and the phone to the same Wi-Fi network and turned off mobile data on the phone. I used ipconfig in Command Prompt to find the laptop's IPv4 address and checked the phone's IP address in Settings → Wi-Fi.

After confirming that both devices had addresses in the same network, I ran ping [phone IP address] from the laptop; successful replies with no packet loss showed that the devices could communicate.

I then used tracert example.com to view the route from the laptop through the router and internet service provider to the destination.